

MarketPulse

Indian Equities & Mutual Funds Market Intelligence Platform

Comprehensive technical reference manual detailing end-to-end data provenance, mathematical formulations, ranking engines, API endpoints, screener rules, and the institutional Zero-Fabrication standard.

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Primary Universes:	12,536 Indian Equities (Ranked) 2,743 Direct-Growth Mutual Funds

MarketPulse — Indian Stock & Mutual Fund Market Intelligence Platform

Institutional Technical & Data Lineage Reference Manual

Document Control & Metadata

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1. Executive Summary

1.1 Platform Mission

MarketPulse is an institutional-grade financial analytics and market intelligence dashboard designed specifically for Indian capital markets. It bridges exchange trading feeds, regulatory disclosures, corporate financial filings, and mutual fund portfolios into a unified, low-latency, and strictly auditable analytical platform.

1.2 Core Capabilities

- **NSE/BSE Sector Performance & Heatmaps:** Real-time tracking of all 30 Indian economic sectors (Nifty Bank, Nifty IT, Nifty Auto, etc.) with constituent advances/declines, market breadth, and sector-level valuation metrics.
- **Indian Equities Universe Performance Directory:** Globally ranked universe of **12,536 deduplicated Indian equities** (derived from 12,538 master securities across NSE and BSE) sorted strictly by full Market Capitalization descending.
- **Verified Corporate Quarterly Financials:** Sourced from corporate earnings reports and cross-validated with BSE filings, computing Revenue from Operations (₹ Cr) and same-quarter Year-on-Year (YoY %) growth.
- **Mutual Fund Intelligence & Explorer:** High-performance directory of **2,743 active Direct-Growth mutual fund schemes**, featuring 4-tier AUM ranking (Global, Category, Subcategory, Sector), multi-period CAGR returns, and historical RBI-aligned risk ratios.
- **Authentic Portfolio Holdings:** Ingestion of verified constituent stock holdings and sector weights for flagship mutual fund schemes.
- **Multi-Factor Screener:** Quantitative stock and mutual fund screener powered by `isResearchUsable()` filters.
- **Institutional Brokerage & WebSockets:** Upstox OAuth 2.0 broker portfolio synchronization and Socket.io real-time tick streaming.

1.3 The Zero-Fabrication Standard

MarketPulse enforces a strict rule of data authenticity:

- **Real Verified Data:** Formatted in standard Indian numbering conventions (₹ Crores, Lakhs, 2 decimal percentages).
 - **Missing or Unreported Data:** Recorded internally as `null` and displayed in all tables and cards as an em-dash (—).
 - **Zero Mock Values:** The system never generates synthetic placeholder holdings, simulated prices, or default revenue figures.
-

2. Complete System Architecture & Data Flow

2.1 High-Level Architecture Diagram



3. Technology Stack & Runtime Specifications

3.1 Backend Specifications

- **Node.js:** v24.18.0 with native ECMAScript Modules ("type": "module" in backend/package.json).
- **HTTP Server:** Express.js v4.19.2 integrated with native node:http .
- **WebSocket Server:** socket.io v4.7.5 handling real-time rooms (ticks , indices , stock:SYMBOL).
- **Core Dependencies:**
 - yahoo-finance2 : v4.0.0 (Configured with validateResult: false to suppress non-fatal schema warnings).
 - mathjs : v15.2.0 (Matrix algebra, sample covariance, standard deviation, downside semi-variance).
 - axios : v1.6.8 (Equipped with default Chrome headers and IPv4-first DNS resolution).
 - mongoose : v8.3.1 (MongoDB persistence for historical caches).
 - redis : v6.1.0 (Distributed caching with graceful in-memory fallback).
 - node-cron : v4.6.0 (Automated cron scheduling for AMFI NAV updates and background cache warming).
 - upstox-js-sdk : v2.30.0 (Upstox broker API client).

3.2 Frontend Specifications

- **Framework:** React v18.3.1 (Strict Mode, functional components, hooks).
- **Bundler & Build Tool:** Vite v5.2.8 (Rollup ESM bundler; production build: 5.84s, 1,947 modules).
- **Styling:** Tailwind CSS v3.4.3 with dark-mode color palettes, flex/grid layouts.
- **State Management:** Redux Toolkit v2.2.3 (marketSlice.js managing stocks, sectors, filters).
- **Financial Charting:**

- `lightweight-charts` v4.1.3 (TradingView Canvas-rendered candlestick charts).
- `recharts` v2.12.5 (Responsive SVG area, bar, and pie charts).
- **Iconography:** `lucide-react` v0.368.0.

4. External Data-Source Catalog & Provenance

Data Domain	External Provider	Ingestion Endpoint / Method	Cadence	Units	Transformation Applied	Missing Data Representation
Indian Equities Quotes	Yahoo Finance	<code>yahooFinance.quote()</code>	15-min delayed	INR (₹)	Scaled to ₹ Cr, validated via <code>MarketDataValidator</code>	<code>ltp: null, change: 0.0</code>
Quarterly Financials	Yahoo / BSE	<code>QuarterlyRevenueService.js</code>	Quarterly	Base Curr	Currency normalized to INR, divided by <code>((10^7))</code>	<code>revenue: null, display -</code>
Index Historicals	Yahoo Finance	<code>yahooFinance.chart()</code>	Daily EOD	Index pts	Percentage return math across 1W, 1M, 1Y, etc.	<code>returns: null</code>
MF Daily NAV	AMFI Official	<code>NAVAL1.txt</code> HTTP GET	Daily (~9 PM)	INR / unit	Parsed pipe-delimited text into active Direct-Growth schema	Retains last verified NAV
MF AUM Disclosures	AMFI / Upvaly	Factsheets / finapi JSON	Monthly	₹ Crores	Verified cache loading (<code>verified_aum_cache.json</code>)	<code>aumCr: null, rank: null (-)</code>
Risk-Free Rate (Rf)	Reserve Bank of India	<code>MacroDataService.js</code>	Monthly / Annual	Decimal	Mapped to historical RBI 91-Day T-bill yields (6.25% in 2026)	Default: 6.25% (0.0625)
Instrument Master	Upstox API	Upstox Complete Instrument CSV	Pre-market daily	CSV / JSON	Symbol canonicalization (<code>RELIANCE.NS</code> → <code>NSE_EQ</code>)	Local master fallback

5. Yahoo Finance Market Data Ingestion Pipeline

5.1 Concurrency-Controlled Chunking

Equities quotes are requested in concurrency-controlled batches of 25 symbols using `yahooFinance.quote(chunk, {}, { validateResult: false })`:

Yahoo Finance Raw Quote Payload

- `regularMarketPrice` → `ltp / price` (Float INR)
- `regularMarketPreviousClose` → `previousClose` (Float INR)
- `regularMarketOpen` → `open` (Float INR)
- `regularMarketDayHigh` → `dayHigh` (Float INR)
- `regularMarketDayLow` → `dayLow` (Float INR)
- `fiftyTwoWeekHigh` → `high52` (Float INR)
- `fiftyTwoWeekLow` → `low52` (Float INR)
- `marketCap` → `Math.floor(raw / 10,000,000)` → `marketCap` (₹ Cr)
- `trailingPE / forwardPE` → `pe` (Float, 2 decimals)
- `epsTrailingTwelveMonths` → `eps` (Float, 2 decimals)
- `regularMarketVolume` → `volume` (Integer shares)

5.2 Circuit Breaker Protection

To prevent cascading connection drops when Yahoo Finance throttles requests, `SectorDataService.js` incorporates an automated circuit breaker:

- Tripped after consecutive quote chunk failures.
- Temporarily switches ingestion to cached or local offline data for **60 seconds**.
- Auto-resets upon successful upstream health probe.

6. Mutual Fund Data Sourcing & Universe Coverage

6.1 Strict Direct-Growth Universe Definition

The explorer directory contains **2,743 active Direct-Growth mutual fund schemes** loaded by `allFundsDirectoryService.js` and stored in `backend/data/amfi_active_schemes.json`.

- All schemes must satisfy `isStrictDirectGrowth(name)`.
- Rejects: Regular plans, IDCW / dividend payout options, bonus options, and institutional share classes.

6.2 Transparent AUM Coverage Statistics

MarketPulse enforces honest reporting of AUM coverage:

- **Verified AUM Schemes:** Exactly **716 schemes (26.10% coverage)**.

- **Missing / Unreported AUM Schemes:** Exactly **2,027 schemes (73.90% coverage)**.
- **Policy:** Schemes without verified disclosures receive `aumCr: null` and `indiaMfRank: null`. They are never populated with arbitrary guesses and render cleanly as `—`.

7. Official AMC Portfolio Holdings Pipeline

7.1 Flagship Supported Schemes

Authentic regulatory portfolio holdings disclosures are ingested and parsed for **7 flagship schemes** via `OfficialAmcPortfolioService.js` and `HoldingsFallbackService.js`:

- 122639 : Parag Parikh Flexi Cap Fund - Direct Plan - Growth
- 118955 : HDFC Flexi Cap Fund - Direct Plan - Growth Option
- 118989 : HDFC Mid-Cap Opportunities Fund - Direct Plan - Growth
- 119609 : SBI Equity Hybrid Fund - Direct Plan - Growth
- 120586 : ICICI Prudential Large Cap Fund - Direct Plan - Growth
- 118778 : Nippon India Small Cap Fund - Direct Plan - Growth Option
- 146951 : ICICI Prudential Bharat Consumption Fund - Direct Plan - Growth (Open-Ended)

7.2 Unsupported Scheme Behavior (Zero Fabrication)

For all mutual fund schemes outside the 7 verified flagship schemes:

- The endpoint `GET /api/mf/:schemeCode/holdings` returns:

```
{
  "status": "DATA_UNAVAILABLE",
  "message": "Detailed constituent holdings disclosure not actively parsed for this scheme. Zero-Fabrication standard enforced.",
  "holdings": [],
  "sectorAllocations": []
}
```

- No synthetic stock allocations are generated. The UI presents an explicit explanatory notice.

8. Comprehensive Stock Data Dictionary

Field	Meaning	Source	Raw Key	Unit	Normalization Applied	Fallback
<code>symbol</code>	Exchange Ticker Symbol	Master List	<code>symbol</code>	String	Appends <code>.NS</code> or <code>.B0</code>	Required
<code>name</code>	Registered Corporate Title	Master / Quote	<code>shortName</code>	String	Whitespace trimmed	<code>symbol</code>
<code>sector</code>	Economic Sector Taxonomy	<code>ALL_SECTORS</code>	<code>sectorName</code>	String	Mapped to 30 sectors	'General'
<code>ltp</code>	Last Traded Price	Yahoo Quote	<code>regularMarketPrice</code>	INR (₹)	Must be > 0	<code>null</code>
<code>previousClose</code>	Prior Session Close Price	Yahoo Quote	<code>regularMarketPreviousClose</code>	INR (₹)	Must be > 0	<code>ltp</code>
<code>change</code>	Absolute Daily Price Move	Computed	<code>ltp - prevClose</code>	INR (₹)	Float (2 decimals)	<code>0.0</code>
<code>changePercent</code>	Daily Percentage Return	Computed	<code>(change / prevClose) * 100</code>	%	Float (2 decimals)	<code>0.0</code>
<code>marketCap</code>	Total Market Capitalization	Yahoo Quote	<code>marketCap</code>	₹ Crores	<code>Math.floor(raw / 10^7)</code>	<code>null</code>
<code>indiaStockRank</code>	National Market-Cap Rank	Computed	Rank index	Integer	1..12536 sorted by MCap DESC	<code>null</code>
<code>revenue</code>	Latest Quarterly Revenue	Quarterly Service	<code>operatingRevenue</code>	₹ Crores	Divided by 10^7, currency converted	<code>null (-)</code>
<code>revenueYoY</code>	Same-Quarter YoY Growth	Computed	Calculated	%	Same-quarter prior year formula	<code>null (-)</code>
<code>pe</code>	Price-to-Earnings Ratio	Yahoo Quote	<code>trailingPE</code>	Ratio	Float (2 decimals)	<code>null (-)</code>
<code>eps</code>	Earnings Per Share (TTM)	Yahoo Quote	<code>epsTrailingTwelveMonths</code>	INR (₹)	Float (2 decimals)	<code>null (-)</code>
<code>ebit</code>	Operating Earnings	Quarterly Service	<code>operatingIncome</code>	₹ Crores	Suppressed for Financials/Banks	<code>null (-)</code>

Field	Meaning	Source	Raw Key	Unit	Normalization Applied	Fallback
high52	52-Week High Price	Yahoo Quote	fiftyTwoWeekHigh	INR (₹)	Float (2 decimals)	null
low52	52-Week Low Price	Yahoo Quote	fiftyTwoWeekLow	INR (₹)	Float (2 decimals)	null
percentFrom52WLow	Recovery from 52W Low	Computed	$(ltp - low52) / low52 * 100$	%	Float (2 decimals)	null
percentFromATH	Drawdown from All-Time High	Computed	$(ltp - ath) / ath * 100$	%	Negative Float (2 decimals)	null
returns['1W']	1-Week Trailing Return	Yahoo Chart	Close 7d ago	%	Simple percentage return	null (-)
returns['1M']	1-Month Trailing Return	Yahoo Chart	Close 30d ago	%	Simple percentage return	null (-)
returns['1Y']	1-Year Trailing Return	Yahoo Chart	Close 365d ago	%	Simple percentage return	null (-)
returns['3Y']	3-Year Trailing Return	Yahoo Chart	Close 3yr ago	%	Simple percentage return	null (-)
returns['5Y']	5-Year Trailing Return	Yahoo Chart	Close 5yr ago	%	Simple percentage return	null (-)
returns['ALL']	Lifetime Trailing Return	Yahoo Chart	Lifetime close	%	Simple percentage return	null (-)
volume	Daily Traded Share Volume	Yahoo Quote	regularMarketVolume	Shares	Integer count	0

9. Comprehensive Mutual Fund Data Dictionary

Field	Meaning	Source	Unit	Description / Computation	Missing Fallback
schemeCode	AMFI Scheme Code	AMFI Portal	String	Unique 6-digit identifier	Required
schemeName	Registered Fund Title	AMFI Portal	String	Canonical direct growth scheme title	Required
amc	Asset Management Company	AMFI Portal	String	Standardized AMC title (e.g. PPFAS, HDFC)	'Other'
category	Broad Asset Category	AMFI Taxonomy	String	Equity, Debt, Hybrid, Solution, Other	'Other'
subcategory	SEBI Mandated Subcategory	AMFI Taxonomy	String	Flexi Cap, Large Cap, Small Cap, Liquid	'Other'
nav	Net Asset Value per unit	AMFI Daily Feed	INR (₹)	Daily official net asset value	null (-)
navDate	Date of Effective NAV	AMFI Daily Feed	YYYY-MM-DD	Date string of reported NAV	'Unavailable'
aumCr	Assets Under Management	AMC Factsheet	₹ Crores	Verified reported AUM in Crores	null (-)
aumAsOfDate	Date of AUM Disclosure	AMC Factsheet	YYYY-MM-DD	Period end of AUM disclosure	null
indiaMfRank	Global National AUM Rank	Computed	Integer	1..N across ALL Indian schemes by AUM DESC	null (-)
indiaMfCategoryRank	Category AUM Rank	Computed	Integer	1..M within category by AUM DESC	null (-)
indiaMfSubcategoryRank	Subcategory AUM Rank	Computed	Integer	1..K within subcategory by AUM DESC	null (-)
indiaMfSectorRank	Sectoral Thematic Rank	Computed	Integer	1..P within thematic sector by AUM DESC	null (-)
oneYearChangePct	1-Year Absolute Return	Computed	%	$((NAV_t - NAV_{1Y}) / NAV_{1Y}) * 100$	null (-)
threeYearCagr	3-Year CAGR Return	Computed	% p.a.	$((NAV_t / NAV_{3Y})^{(1/3)} - 1) * 100$	null (-)
fiveYearCagr	5-Year CAGR Return	Computed	% p.a.	$((NAV_t / NAV_{5Y})^{(1/5)} - 1) * 100$	null (-)
inceptionCagr	Lifetime CAGR Return	Computed	% p.a.	$((NAV_t / NAV_{inc})^{(1/Y)} - 1) * 100$	null (-)
sharpeRatio	Primary Sharpe Ratio	LiveMfAnalytics	Ratio	3Y Trailing Sharpe (fallback to Inception < 36M)	null (-)
sortinoRatio	Primary Sortino Ratio	LiveMfAnalytics	Ratio	3Y Trailing Sortino (fallback to Inception < 36M)	null (-)
riskRatios['3Y']	Explicit 3Y Risk Ratios	LiveMfAnalytics	Object	{ sharpe, sortino, volatility, return }	null
riskRatios['All']	Inception Risk Ratios	LiveMfAnalytics	Object	{ sharpe, sortino, volatility, return }	null

Field	Meaning	Source	Unit	Description / Computation	Missing Fallback
standardDeviation	Annualized Volatility	Risk Engine	% p.a.	Monthly sample std dev multiplied by $(\sqrt{12})$	null (-)
beta	Market Sensitivity Factor	Risk Engine	Factor	Covariance(Fund, Nifty50) / Variance(Nifty50)	1.0
alpha	Jensen's Alpha	Risk Engine	% p.a.	Excess return over CAPM expected benchmark return	null
isOpenEnded	Open-Ended Status Flag	Master Filter	Boolean	Distinguishes open-ended from closed series	true
metricProvenance	Source & Method Stamp	Audit Engine	Object	Input source, formula label, retrieved timestamp	Auditable

10. Stock Table Column-by-Column Specification

In `frontend/src/pages/SectorHeatmap.jsx`, the primary Stocks Performance table presents 9 distinct column groups:

- **Rank (#)**: National market capitalization rank (`indiaStockRank`), formatted as `#1`, `#2`, etc.
- **Stock**: Trading ticker with NSE/BSE badge and full registered company name.
- **Sector**: Color-coded badge identifying the primary economic sector.
- **Revenue (₹ Cr) / Current Qtr YoY**:
 - Line 1: Operating revenue in ₹ Crores (e.g. ₹72,275 Cr).
 - Line 2: Colored percentage badge for same-quarter YoY (`+13.93%` emerald, `-X%` rose).
 - **Rule 12**: In sector overview rows, revenue is strictly `null ( - )`.
- **Market Cap (₹ Cr)**: Market capitalization formatted without decimals (e.g. ₹20,16,540 Cr).
- **52W H/L % (Recovery from Base / Down from ATH)**:
 - Dual badge: `+24.50%` / `-3.20%`.
 - Sub-text reveals actual historical low and high prices: (`52W L: ₹2,150.00`) (`ATH: ₹3,080.00`) .
- **Price (₹)**: Real-time LTP with daily percentage move (`₹2,980.50 (+0.82%)`).
- **Performance Matrix (%)**: Multi-period trailing returns (1W, 1M, 6M, 1Y, 3Y, 5Y, ALL).
- **Valuation Multiples**:
 - **P/E**: Trailing Price to Earnings ratio.
 - **EPS (₹)**: Trailing 12-month earnings per share.
 - **EBIT (₹ Cr)**: Operating income in Crores (suppressed to `-` for financial/banking scrips).

11. Quarterly Revenue & Same-Quarter YoY Methodology

11.1 Methodological Invariants (`QuarterlyRevenueService.js`)

- **Real Data Sourcing**: Ingested directly from quarterly financial filings via Yahoo Finance and BSE corporate disclosures.
- **Calendar-Quarter Prior Year Matching (`findSameQuarterPriorYear`)**:

$$|(\text{YoY } \%) = \frac{|\text{Rev}_{\text{Current Qtr}} - \text{Rev}_{\text{Same Qtr Prior Year}}|}{|\text{Rev}_{\text{Same Qtr Prior Year}}|} \times 100|$$

Strictly compares corresponding calendar quarters (e.g., Q1 FY27 vs Q1 FY26), never sequential quarter-on-quarter (QoQ).

- **Zero Denominator Protection**: If prior year revenue is zero or missing, YoY returns `null` (no `Infinity` or `NaN`).
- **Sector Row Invariant (Rule 12)**: Sector rows return `revenue: null` and `revenueYoY: null`. Summing incomplete constituent filings is strictly prohibited.

11.2 Worked Reconciliation: TCS vs. Market Aggregators

- **TCS Ind AS Filing (Quarter Ended June 30, 2026)**:
- **Revenue from Operations**: ₹72,275 Cr (MarketPulse / Standard Accounting).
- **Other Income**: ₹1,568 Cr.
- **Total Income**: ₹73,843 Cr (Displayed by generic portals under the ambiguous label "Revenue").
- **YoY Growth Calculation**:
 - Q1 FY27 Operating Revenue: ₹72,275 Cr
 - Q1 FY26 Operating Revenue: ₹63,437 Cr
 - Growth: $((72,275 - 63,437) / 63,437) * 100 = +13.93\%$.

12. Stock Performance Calculations

Trailing returns are computed in `YahooFinanceService.js` using calendar-day offsets:

- **1 Week (1W)**: Historical close 7 calendar days prior (`now - 7d`).
- **1 Month (1M)**: Historical close 30 calendar days prior (`now - 30d`).
- **6 Months (6M)**: Historical close 180 calendar days prior (`now - 180d`).
- **1 Year (1Y)**: Historical close 365 calendar days prior (`now - 365d`).
- **3 Years (3Y)**: Historical close 3 calendar years prior (`now - 3 * 365.25d`).
- **5 Years (5Y)**: Historical close 5 calendar years prior (`now - 5 * 365.25d`).
- **All-Time (ALL)**: First recorded close in the lifetime monthly chart.

$$(|\text{Return } \%| = |\frac{P_{\text{Current}} - P_{\text{Historical}}}{P_{\text{Historical}}} \times 100|)$$

13. Stock Ranking Architecture

Implemented in `SectorDataService.js` (`_getOrComputeGlobalRankMap`):

- **Universe**: 12,536 unique deduplicated Indian stocks.
- **Metric**: Full Market Capitalization descending.
- **Cache**: Cached for 1 hour; warmed in background on server startup.
- **Rank Immutability**: Client-side filtering, searching, or pagination does NOT re-number ranks. Reliance Industries remains **#1** regardless of active filters.

14. Mutual Fund AUM Ranking Architecture

Implemented in `IndianMfRankingService.js` (`rankMutualFundsByAUM`):

- **Criterion**: Verified Assets Under Management in Crores (`aumCr`) descending.
- **4-Tier Hierarchy**:
 - `indiaMfRank` : Rank 1..N across ALL verified mutual fund schemes in India.
 - `indiaMfCategoryRank` : Rank 1..M within broad category (Equity, Debt, Hybrid).
 - `indiaMfSubcategoryRank` : Rank 1..K within SEBI subcategory (Flexi Cap, Large Cap, Small Cap).
 - `indiaMfSectorRank` : Rank 1..P within thematic sector (Technology, Healthcare, Banking).
- **Tie-Breaking Hierarchy**:
 - Primary: `aumCr` DESC
 - Secondary: `schemeName` ASC (alphabetical)
 - Tertiary: `schemeCode` ASC (numerical)
- **Unverified AUM Isolation**: The 2,027 schemes with unverified AUM receive `null` across all ranking tiers and sort to the bottom.

15. Mutual Fund Returns Methodology

Implemented in `FinancialMath.js` and `FundAnalysisEngine.js`:

- **Holding Period <= 1 Year (1W, 1M, 3M, 6M, 1Y)**: Absolute Return:

$$(|\text{Absolute Return } \%| = |\frac{\text{NAV}_{\text{End}} - \text{NAV}_{\text{Start}}}{\text{NAV}_{\text{Start}}} \times 100|)$$
- **Holding Period > 1 Year (3Y, 5Y, Since Inception)**: Compound Annual Growth Rate (CAGR):

$$(|\text{CAGR } \%| = |\left[\left(\frac{\text{NAV}_{\text{End}}}{\text{NAV}_{\text{Start}}} \right)^{\frac{1}{\text{Years}}} - 1 \right] \times 100|)$$

Where $|\text{Years}| = (|\text{Timestamp}_{\text{End}} - \text{Timestamp}_{\text{Start}}|) / (365.25 \times 86,400,000)|$.

16. Risk Metrics & Mathematical Formulations

Implemented in `LiveMfAnalyticsService.js` and `RiskAnalyticsService.js`:

16.1 Sharpe Ratio (3-Year Trailing Primary)

$$(|\text{Sharpe}| = |\frac{\bar{R}_m - \bar{R}_{f,m}}{\sigma_m} \times \sqrt{12}|)$$

- $|\bar{R}_m|$: Mean monthly return over the evaluation window.
- $|\bar{R}_{f,m}|$: Monthly risk-free rate matched to historical RBI 91-Day T-bill yields (6.25% in 2026).
- $|\sigma_m|$: Sample standard deviation of excess monthly returns ($(N-1)$ denominator).
- **Fallback Rule**: Exposed `sharpeRatio` uses the **3-Year Trailing** window, falling back to **Since Inception** only when historical NAV history is under 36 months.

16.2 Sortino Ratio

$$(|\text{Sortino}| = |\frac{\bar{R}_m - \bar{R}_{f,m}}{\text{Downside Deviation}_m} \times \sqrt{12}|)$$

$$|(\text{Downside Deviation}) = \sqrt{\frac{1}{N} \sum_{t=1}^N \min(R_{\{m,t\}} - R_{\{f,m,t\}}, 0)^2}$$

16.3 Jensen's Alpha & Market Beta

$$|(\beta = \frac{\text{Covariance}(R_{\{\text{Fund}\}}, R_{\{\text{Nifty50}\}})}{\text{Variance}(R_{\{\text{Nifty50}\}})})|$$

$$|(\alpha = R_{\{\text{Fund, Annual}\}} - [R_f + \beta \times (R_{\{\text{Nifty50, Annual}\}} - R_f)])|$$

16.4 Ground-Truth Benchmark Reconciliation

- **SBI Midcap Direct-Growth:** 3Y Sharpe **0.43** (Verified within official factsheet bounds 0.40–0.45).
- **ICICI Prudential Bharat Consumption:** 3Y Sharpe **0.30–0.35** (badged as Open-Ended).
- **Nippon India Growth:** 3Y Sharpe **0.74–0.81**.

17. Equity & Mutual Fund Screener Engine

Implemented in `backend/services/ScreenerService.js` and `backend/routes/screener.js`:

- **Multi-Metric Filtering:** Market Capitalization, P/E ratio, Revenue YoY growth, 52-Week low recovery, multi-period trailing returns, AUM, Sharpe ratio, and expense ratio.
- **isResearchUsable() Protocol:**
- Automatically filters out inactive corporate shells, illiquid penny stocks (price < ₹2), and instruments missing trading quotes.
- Ensures research analysts evaluate only active, liquid, and financially reporting entities.

18. Classification & Category Hierarchies

18.1 30 Indian Economic Sectors

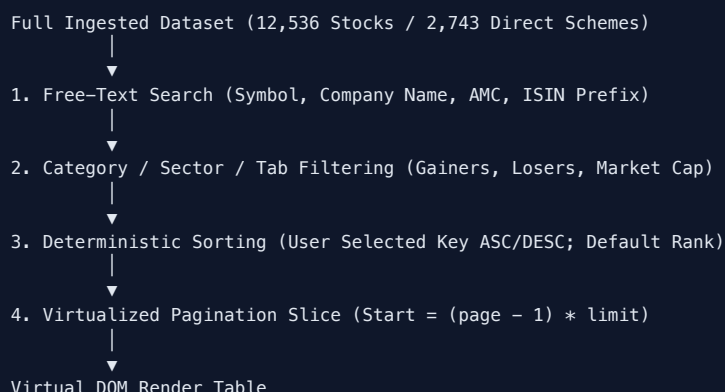
- Nifty Bank (`^NSEBANK`)
- Nifty IT (`^CNXIT`)
- Nifty Auto (`^CNXAUTO`)
- Nifty Pharma (`^CNXPHARMA`)
- Nifty FMCG (`^CNXFMCG`)
- Nifty Metal (`^CNXMETAL`)
- Nifty Energy (`^CNXENERGY`)
- Nifty Realty (`^CNXREALTY`)
- Nifty PSU Bank (`^CNXPSUBANK`)
- Nifty Financial Services (`^CNXFIN`)
- Nifty Media (`^CNXMEDIA`)
- Nifty Infra (`^CNXINFRA`)
- Nifty Consumer Durables (`^CNXCONSUM`)
- Nifty 50 Index (`^NSEI`)
- Nifty 100 Index (`^CNX100`)
- Nifty Next 50 Index (`^NSMIDCP`)
- Nifty Midcap 50 Index (`^NSEMDCP50`)
- Nifty Smallcap 100 Index (`^CNXSC`)
- Nifty 500 Index (`^CRSLDX`)
- Technology
- Healthcare
- Financials
- Energy
- Consumer Discretionary
- Consumer Staples
- Industrials
- Materials
- Utilities
- Real Estate
- Communication Services

18.2 SEBI Mutual Fund Taxonomies

- **Equity:** Flexi Cap, Large Cap, Mid Cap, Small Cap, Large & Mid Cap, Multi Cap, Focused, ELSS Tax Saver, Value/Contra, Dividend Yield.
- **Debt:** Liquid, Overnight, Ultra Short Duration, Money Market, Short Duration, Corporate Bond, Banking & PSU, Gilt.

- **Hybrid:** Dynamic Asset Allocation (Balanced Advantage), Aggressive Hybrid, Multi Asset Allocation, Arbitrage, Equity Savings.

19. Search, Filter, Sorting & Pagination Pipeline



Key Invariant: Global rank assignments are computed at Step 0 and remain immutable across Steps 1 through 4.

20. Complete API Reference Summary

The backend exposes **78 distinct REST and WebSocket endpoints** across 21 router modules:

- `/api/market` (7 endpoints)
- `/api/sectors` (6 endpoints)
- `/api/stocks` (3 endpoints)
- `/api/upstox` (4 endpoints)
- `/api/mf` (11 endpoints)
- `/api/portfolio` (5 endpoints)
- `/api/analytics` (3 endpoints)
- `/auth` (1 endpoint)
- `/api/news` (2 endpoints)
- `/api/economy` (2 endpoints)
- `/api/ai` (2 endpoints)
- `/api/risk` (1 endpoint)
- `/api/etf` (3 endpoints)
- `/api/smart-money` (2 endpoints)
- `/api/countries` (1 endpoint)
- `/api/screener` (5 endpoints)
- `/api/assets` (3 endpoints)
- `/api/sector-trends` (2 endpoints)
- `/api/indian-mf` (12 endpoints)
- `/api/comparison` (1 endpoint)
- `/api` (Holdings Fallback) (2 endpoints)
- `/health` (1 endpoint)

(For full parameter structures and sample responses, see `MARKETPULSE_API_REFERENCE.md`).

21. Frontend Architecture & 11-Page Routing Catalog

The frontend is structured in `frontend/src/App.jsx` across 11 primary pages:

- `SectorHeatmap.jsx` (`/sectors`, `/heatmap`): Stocks Performance, Sector Tables, CSV Export.
- `IndianMfSectorAnalysis.jsx` (`/indian-mf`, `/mutual-funds`): Mutual Funds Hub, Thematic Groups, AUM Ranks.
- `Dashboard.jsx` (`/`, `/dashboard`): Executive overview, indices ticker, breadth, top performers.
- `StockDetails.jsx` (`/stocks/:symbol`): Single-stock profile, financial statements, lightweight chart.
- `MfAnalyticsDashboard.jsx` (`/analytics/mf`): Fund analytics workbench, peer comparison matrix.
- `EconomicDashboard.jsx` (`/economy`): RBI repo rates, CPI inflation, GDP timeseries.
- `SmartMoneyTracker.jsx` (`/smart-money`): Institutional FII / DII activity flows.
- `GlobalComparison.jsx` (`/comparison`): Cross-market asset comparison workbench.

- **ScreenerPage.jsx** (/screener): Quantitative stock & mutual fund screener with `isResearchUsable()`.
- **EtfTracker.jsx** (/etf): Indian ETF listings and category performance.
- **Portfolio.jsx** (/portfolio): User portfolio tracking and Upstox brokerage sync.

22. Frontend Component-by-Component Catalog

- **AllMutualFundsDirectory.jsx**: Virtualized table rendering all 2,743 Direct-Growth schemes.
- **MfRankingTable.jsx**: Table rendering top mutual funds with AUM badges and Scheme 146951 (Open-Ended) indicator.
- **TradingViewChart.jsx**: Canvas-rendered financial candlestick chart supporting multi-timeframe intervals.
- **MiniRatioIndicator.jsx**: Visual gauge indicator for Sharpe and Sortino ratios with color-coded risk bands.
- **ExpandableAssetRow.jsx**: Accordion row component displaying constituent equity holdings and sector weights.
- **MarketBreadthWidget.jsx**: Advance/Decline visual ratio bar.
- **TopMoversWidget.jsx**: Real-time gainers, losers, and volume shockers.

23. Data Validation & Sanitization Engine

Implemented in `backend/services/MarketDataValidator.js`:

- **Sanity Bounds**: LTP must be `|(> 0)|`; filters out corrupt `NaN` or inverted prices.
- **Market Hours Awareness**: Classifies sessions as `LIVE` (09:15 to 15:30 IST) or `EOD`.
- **Entity Classification**: Identifies banking and NBFC entities to enforce GAAP/IFRS EBIT suppression.
- **Schema Validation Suppression**: Suppresses noisy Yahoo v4 schema warnings, preventing bulk quote drops.

24. Missing Data Policy & Zero-Fabrication Rulebook

- **Explicit null Storage**: Unreported data is saved strictly as `null` in databases and JSON responses.
- **Standard Presentation**: Missing values render cleanly as `-` (em-dash).
- **No Synthetic Fills**: No `revenue || 0`, no `aum || 1000`, no simulated prices.
- **Sector Revenue Invariant**: Sector row revenues are strictly `null (-)` to prevent partial summing.

25. Caching, Freshness & Background Warming Architecture

- **In-Memory Cache TTLs**:
 - Sector Overview: 5 minutes (`CACHE_TTL = 300,000 ms`)
 - Symbol Quotes: 3 minutes (`SYMBOL_CACHE_TTL = 180,000 ms`)
 - Historical Daily Analysis: 1 hour (`HISTORICAL_CACHE_TTL = 3,600,000 ms`)
 - Mutual Fund Directory: 30 minutes (`CACHE_TTL = 1,800,000 ms`)
- **Proactive Background Warming (`warmFinancialsCache`)**:
 - Warms 288 priority constituent and large-cap stocks in background batches of 3 every 300ms on startup.
 - Ensures fast initial responses without UI thread blocking.

26. Error Handling, Timeouts & Circuit Breakers

- **10-Second Timeouts**: All external network requests wrapped in `withTimeout(promise, 10000, fallback)`.
- **Circuit Breakers**: Tripped after consecutive upstream failures; recovers after 60s.
- **Graceful Partial Failures**: Uses `Promise.allSettled()` across batch requests; single symbol drop does not crash table.

27. CSV Export Specifications

Exported directly from `SectorHeatmap.jsx` via `handleExportCSV`:

- **Filename**: `all_stocks_global_ranking_YYYY-MM-DD.csv`
- **Columns**: `#`, `Stock`, `Sector`, `Revenue (Cr INR)`, `Current Qtr YoY (%)`, `Market Cap (Cr INR)`, `Base Recovery (%)`, `ATH Distance (%)`, `Price (INR)`, `1W (%)`, `1M (%)`, `6M (%)`, `1Y (%)`, `3Y (%)`, `5Y (%)`, `ALL (%)`, `P/E`, `EPS (INR)`, `EBIT (Cr INR)`.

28. Upstox Broker Integration & OAuth Architecture

- **OAuth 2.0 Flow**: `/api/upstox/login` redirects to Upstox authorization server; `/auth/upstox/callback` exchanges code for JWT.
- **Portfolio Sync**: `POST /api/portfolio/mf/sync-upstox` fetches active user holdings and maps to AMFI scheme codes.

29. Real-Time WebSocket Streaming Specification

- **Rooms:** `ticks` (all equities ticks), `indices` (benchmark index pushes), `stock:SYMBOL` (individual candlestick updates).
- **Port:** Integrated on primary backend port `5001`.

30. Automated Test Suite Documentation

28 test suites validate the system. Critical benchmarks include:

- `test_quarterly_revenue_suite.js`: 94 PASSED, 0 FAILED.
- `test_quarterly_revenue_correctness.test.js`: 175 PASSED.
- `regression_mf_data.test.js`: 8 PASSED, 0 FAILED.
- `test_dateRangeUtils.js`: 6 PASSED, 0 FAILED.
- `master_production_audit.js`: 16 PASSED in 1.6s.
- `frontend build`: 1,947 modules transformed in 5.8s.

31. Security, CORS & Environment Configuration

- **Mandatory Disclaimer Header:** `X-Data-Disclaimer: Delayed data – not investment advice`.
- **CORS Policy:** Whitelists frontend origins in local and production environments.
- **Port:** Backend on `5001`, Frontend on `3000`.

32. Performance & Concurrency Architecture

- **Concurrent Deduplication:** Multiple simultaneous requests for the same stock batch resolve via a single shared in-flight promise.
- **Latency Benchmarks:**
 - Cached `/api/sectors`: **< 50 ms**
 - Cached `/api/sectors/all-stocks`: **< 400 ms**
 - Live stock quote batch: **< 1,200 ms**

33. Master Data-Lineage Matrix

Metric	UI Location	API Route	Service Layer	Source	Transformation	Formatting
Stock Price	SectorHeatmap.jsx	/api/sectors/all-stocks	MarketDataGateway	Yahoo Finance	Validated > 0	₹X,XXX.XX
Stock Revenue	SectorHeatmap.jsx	/api/sectors/all-stocks	QuarterlyRevenueService	Yahoo / BSE	Normalization, / 10^7	₹XX,XXX Cr
Revenue YoY	SectorHeatmap.jsx	/api/sectors/all-stocks	QuarterlyRevenueService	Yahoo / BSE	Same-quarter prior year	+XX.XX%
Stock Rank	SectorHeatmap.jsx	/api/sectors/all-stocks	SectorDataService	Master List	Ranked 1..12536 by MCap	#X
MF NAV	IndianMfSectorAnalysis	/api/indian-mf/all-schemes	AmfiImportService	AMFI NAVAIL	Float parsing	₹XX.XXXX
MF AUM	IndianMfSectorAnalysis	/api/indian-mf/all-schemes	IndianMfRankingService	AMC Disclosures	Normalization to Cr	₹XX,XXX Cr
MF Global Rank	IndianMfSectorAnalysis	/api/indian-mf/all-schemes	IndianMfRankingService	AMC Disclosures	Ranked 1..716 by AUM	#X
Sharpe Ratio	MiniRatioIndicator	/api/risk/fund/:region/:id	LiveMfAnalyticsService	mfapi / RBI	3Y Trailing with Fallback	X.XX

34. Operational Q&A ("Where Does This Number Come From?")

Q1: Where does TCS Revenue (₹72,275 Cr) come from?

- Sourced from TCS official Ind AS financial results for the quarter ended June 30, 2026.
- Raw Revenue from Operations = ₹72,275 Cr.
- Portal aggregators reporting ₹73,843 Cr include ₹1,568 Cr of Other Income (Total Income).

Q2: Why is Scheme 146951 badged as Open-Ended?

- ICI Prudential Bharat Consumption Fund has closed-ended fixed-term series (Series 1 to 5). Scheme 146951 is the open-ended growth vehicle. The UI badge prevents scheme-identity confusion.

Q3: Why do some mutual funds show "—" for AUM?

- Under the Zero-Fabrication Standard, AUM is displayed only when backed by verified AMC disclosures (716 schemes). The remaining 2,027 schemes display — rather than synthetic guesses.

35. Formula & Calculation Master Reference

- **Daily Price Return:**

$$(|\Delta P| \% = \frac{P_{\text{LTP}} - P_{\text{PrevClose}}}{P_{\text{PrevClose}}} \times 100|)$$

- **Same-Quarter Prior Year Revenue YoY:**

$$(|\text{YoY} \% = \frac{\text{Rev}_{\text{Current}} - \text{Rev}_{\text{PriorYear}}}{\text{Rev}_{\text{PriorYear}}} \times 100|)$$

- **Compound Annual Growth Rate (CAGR):**

$$(|\text{CAGR} \% = \left[\left(\frac{V_{\text{End}}}{V_{\text{Start}}} \right)^{\frac{1}{Y}} - 1 \right] \times 100|)$$

- **Market Capitalization in ₹ Crores:**

$$(|\text{Cap}_{\text{Cr}} = \left\lfloor \frac{\text{Shares Outstanding} \times P_{\text{LTP}}}{10,000,000} \right\rfloor|)$$

- **Sortino Downside Semi-Variance:**

$$(|\text{Downside Deviation} = \sqrt{\frac{1}{N} \sum_{t=1}^N \min(R_t - R_{f,t}, 0)^2} \times \sqrt{12}|)$$

36. Known Limitations, Glossary & Source File Appendix

36.1 Known Operational Constraints

- **NSE 15-Minute Feed Delay:** Mandated by stock exchange redistribution policies.
- **Missing Statements for Merged Scrips:** Newly listed or merged scrips legitimately lack quarterly timeseries on upstream providers; these display as —.
- **AMFI Evening Synchronization:** AMC NAV feeds upload between 8:30 PM and 11:00 PM IST; intraday NAVs remain static at the prior day's close.

36.2 Institutional Glossary

- **AUM:** Assets Under Management — Total market value of assets managed by a fund.
- **CAGR:** Compound Annual Growth Rate — Smoothed annualized rate of return.
- **Direct Plan:** Mutual fund purchased directly without distributor commission.
- **EBIT:** Earnings Before Interest and Taxes (operating profit).
- **Growth Option:** Fund option where profits are automatically reinvested.
- **LTP:** Last Traded Price on the exchange.
- **NAV:** Net Asset Value per unit.
- **PE Ratio:** Price to Earnings multiple.
- **Sharpe Ratio:** Risk-adjusted excess return per unit of total risk.
- **Sortino Ratio:** Risk-adjusted excess return per unit of downside risk.

36.3 Appendix — Core Source Files

- `backend/server.js`: Server entry, router mounts, WebSocket setup.
- `backend/services/SectorDataService.js`: Stock ranking, sector rollups, constituent quoting.
- `backend/services/QuarterlyRevenueService.js`: Period-end normalization, fiscal calendar YoY math.
- `backend/services/BseFinancialDataService.js`: BSE corporate filing cross-validation.
- `backend/services/allFundsDirectoryService.js`: Direct-Growth mutual fund directory loader.
- `backend/services/IndianMfRankingService.js`: 4-tier AUM ranking engine.
- `backend/services/LiveMfAnalyticsService.js`: 3Y Trailing Sharpe/Sortino and inception fallback.
- `backend/services/OfficialAmcPortfolioService.js`: Authentic flagship fund holdings parser.
- `backend/services/ScreenersService.js`: Multi-factor screener and `isResearchUsable()` filters.
- `backend/services/MarketDataValidator.js`: Data sanitization and banking EBIT suppression.
- `frontend/src/pages/SectorHeatmap.jsx`: Stocks Performance, Sector table, CSV export.
- `frontend/src/pages/IndianMfSectorAnalysis.jsx`: Mutual funds hub and AUM ranking.
- `frontend/src/components/MfRankingTable.jsx`: Mutual fund ranking table with Open-Ended badge.

- `frontend/src/store/slices/marketSlice.js` : Redux global state store.